

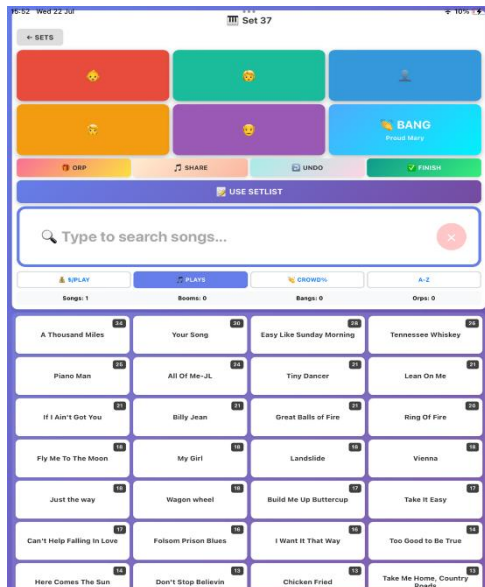
The Stowaway Piano Player Data Project

-By Samuel Martin

I've been a piano entertainer for most of my working life. The last 8 years have been mainly dedicated to playing piano and singing in piano bars on Royal Caribbean ships. The contract I'm on now is a different role, travelling piano man or 'stowaway'. The job involves playing piano all over the ship, doing pop-up performances and playing to whoever is walking by, essentially busking on a cruise ship. I'll be on the Utopia of the Seas for 13 weeks. The most profitable sets I do in terms of tips are in the elevator, yes that's right, I play sets in the elevator! People love these performances; they get very excited. These sets provide a unique opportunity for data collection and analyses. People are getting on the elevator, staying for a few minutes (sometimes more), and then they get off. I have about thirty seconds to play something that gets their attention, to get them singing, laughing or just enjoying the ride! Sometimes playing a quick chorus of something I suspect they know and like works, sometimes just playing a showpiece on the piano or improvising some jazz works, sometimes saying something silly, sometimes playing something the kids will like works, sometimes just playing whatever I feel like. Another great thing about the elevator sets is that people tip, this ship is out of Florida where I know from experience people are generous. As I'm interested in data analytics it gave me the idea to turn this job into a project.



The questions I'll be asking, does song choice matter, does crowd reaction correlate to tips earned. I'll also be using any other insights to try to increase the crowd reaction I get in the elevator and increase the amount of tips I get.



The first problem to solve was the method in which I would collect data. For good quality data I need to track when a song gets a tip and when a song gets a crowd response. This was the biggest challenge. My first attempts failed. I tried note taking, I tried taking an audio recording of the script and extracting songs from that with a code word I would say out loud that meant a tip was received. Those methods proved unworkable. I ended up designing a simple app which I enlisted Claude to build. In the app I start a set, this records the start time, then a song is selected from a list and while the song is 'active' one can

press a button upon receiving a tip and the tip is recorded and attached to the song. the same for crowd reaction. It's a simple system that requires me to select a song before I play it and then simply attach a crowd reaction or a tip to that song. Later iterations of the app included the approximate age of the guest tipping. I was interested to do a generational analysis – what songs do age brackets tip'. At the end of the set the set is closed and the time is recorded. Then there are some other values entered, total earned in tips during the set is recorded after counting up, the location of where the set is and what port we were in that day, the subjective energy level is also tracked along with footfall level. All of these were designed to gain any insights at the end of the experiment.

This is a picture of the app to the left. Songs are selected from the list and the buttons at the top record tips or crowd reaction events attached to that song and are exported in the data.

I then ran the app for every set I played for three weeks. I'd set up in the elevator and start a set, track every song I played. How many tips it earned, whether or not I got people singing along- I landed on the rule that if more than one person sings along, the song event earns a crowd reaction in the data.

The data from the experiment is presented below. I ran all these charts initially in SQL as script which I have published on my GitHub. Given the findings and insights were fairly basic I decided against using Power BI and used the results of my SQL scripts to generate some simple charts within Claude.

THE DATASET

22 days · 33 sets · 42.4 hours of playing.

THE STUDY

Sets played	33
Days played	18
Experiment length	22 days
Total hours performed	42.4
Songs played	1,047
Unique songs drawn on	202

EARNINGS

Total cash	\$3,138
Total tips	507
– Tagged to a song	375
– Orphan (song-independent)	132 (26%)

AVERAGES

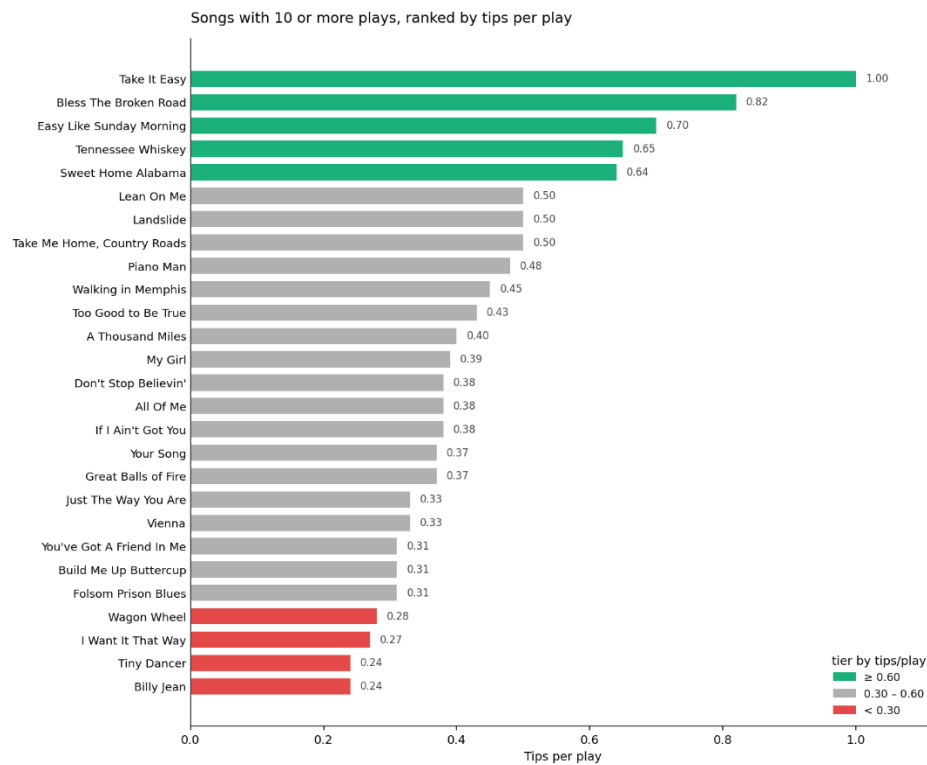
Cash per hour played	\$74
Cash per set	\$95
Cash per song	\$3.00
Tips per set	15.4
Songs per set	31.7
Songs per hour	24.7

BASELINES

Tips per song (all tips)	0.48
Tips per song (tagged only)	0.36

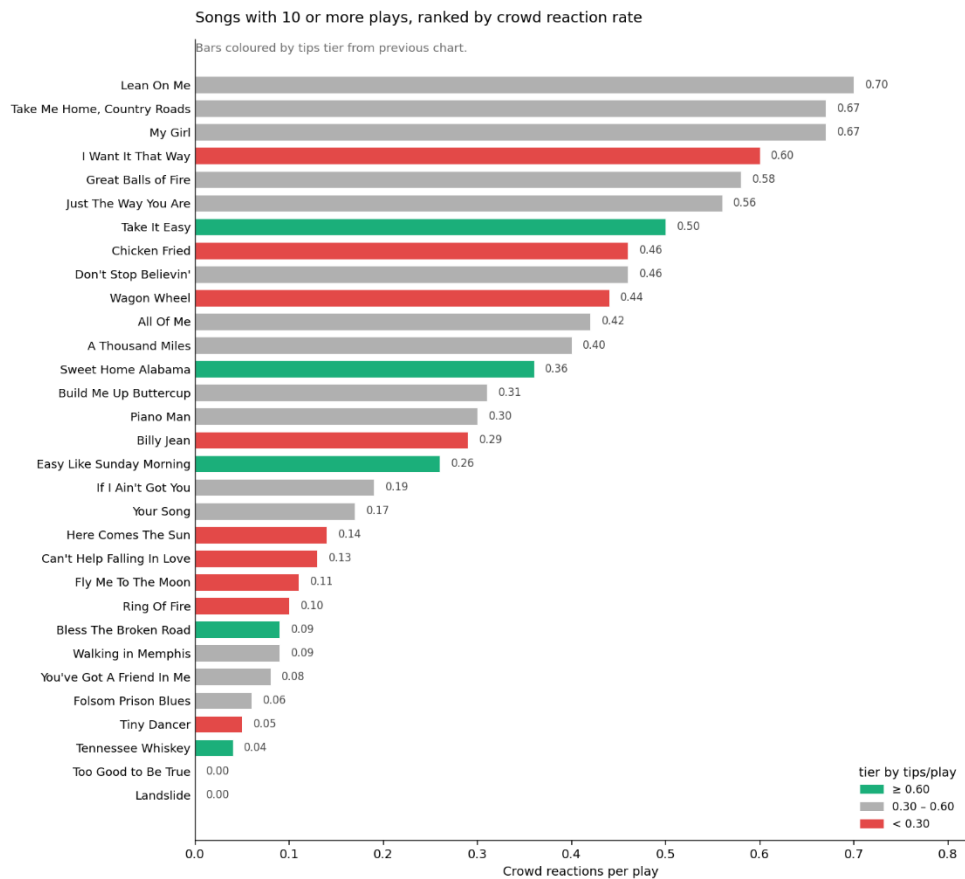
Songs ranked by tips per play

27 songs with 10 or more plays. Green tier earns 0.60+ tips per play, red under 0.30. Baseline for a random play was 0.36. Take It Easy sits at the top with 14 tips in 14 plays; against baseline, that's very unlikely to be chance ($p=0.0007$). The tier the song sits in is more reliable than its exact rate.



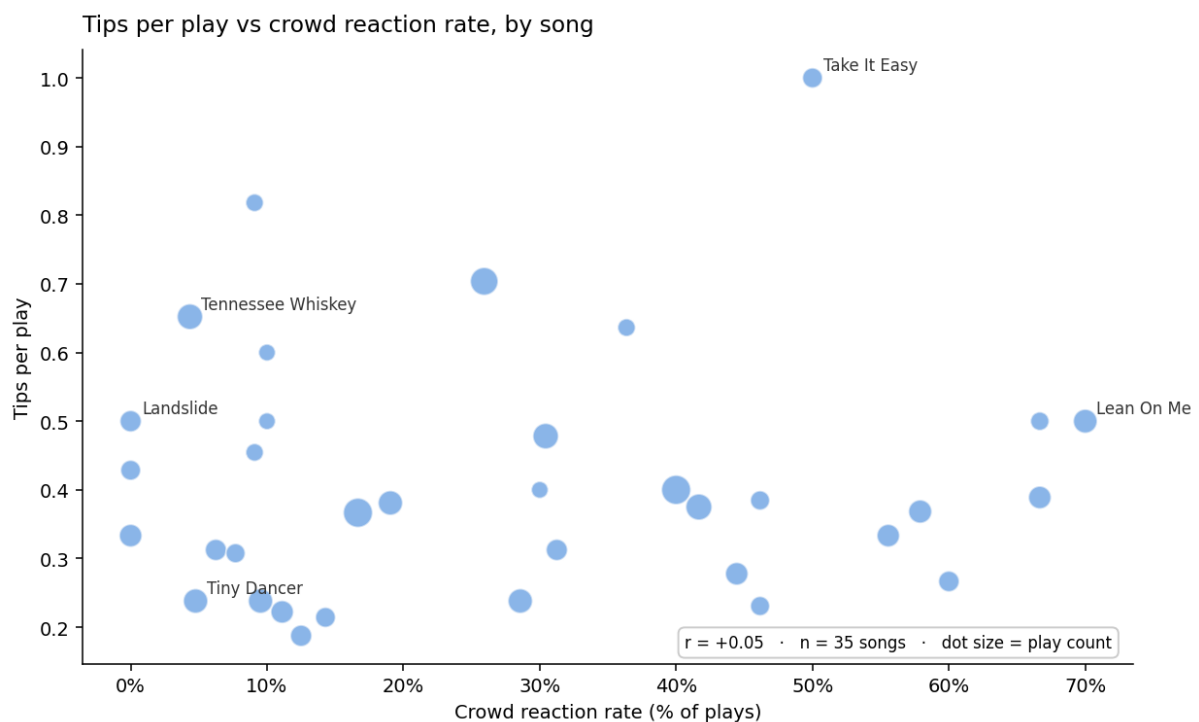
Songs ranked by crowd reaction rate

The same songs re-ranked by how often two or more people sang along. Bars are still coloured by their tips tier — so green scattered through the middle and red near the top means the two rankings don't agree. Lean On Me draws the biggest response but earns mid-tier tips. Tennessee Whiskey and Bless The Broken Road earn top-tier tips almost silently.



Tips per play vs crowd reaction rate, by song

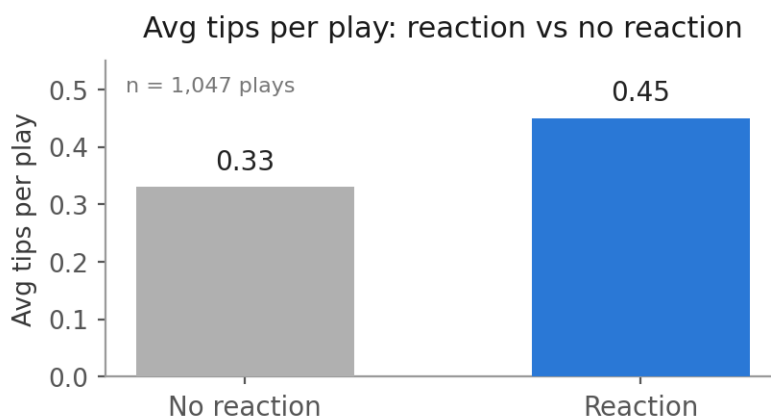
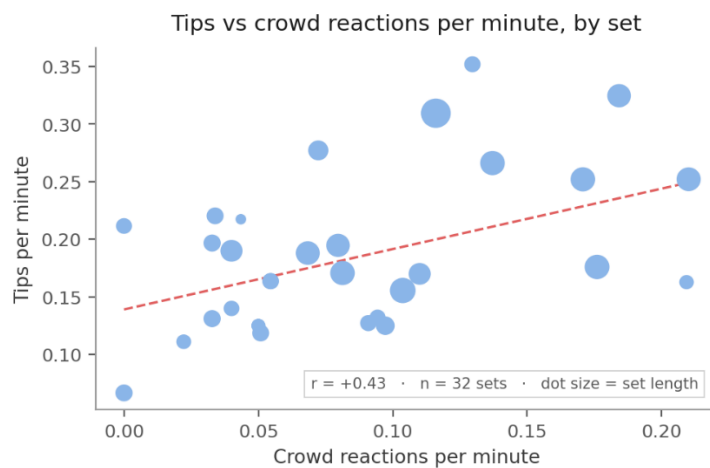
Each dot is one song, sized by play count. If crowd reaction predicted earnings the cloud would slope. It doesn't ($r=0.05$ across 35 songs). A play that gets a reaction does tend to earn more in the moment (0.45 vs 0.33 average tips) — but a song's tendency to *produce* reactions doesn't make it a top earner.



So although there are some reliably good earners and bangers, we can't depend on any one song's singalongability to reliably increase tip count. The top earners are likely a result of me playing the song to the right crowd, my intuition comes into play here and is discussed later in the analysis.

Tips vs crowd reactions per minute, by set

Each dot is one completed set, sized by set length. Unlike the song-level picture, this one does slope: sets with more crowd reactions per minute earned more tips per minute ($r=0.43$, $p=0.013$ across 32 sets). An engaged room tips more — the reaction just isn't something any particular song can reliably produce. This is correlation, not causation: a lively, generous crowd likely drives both the singing and the tipping.

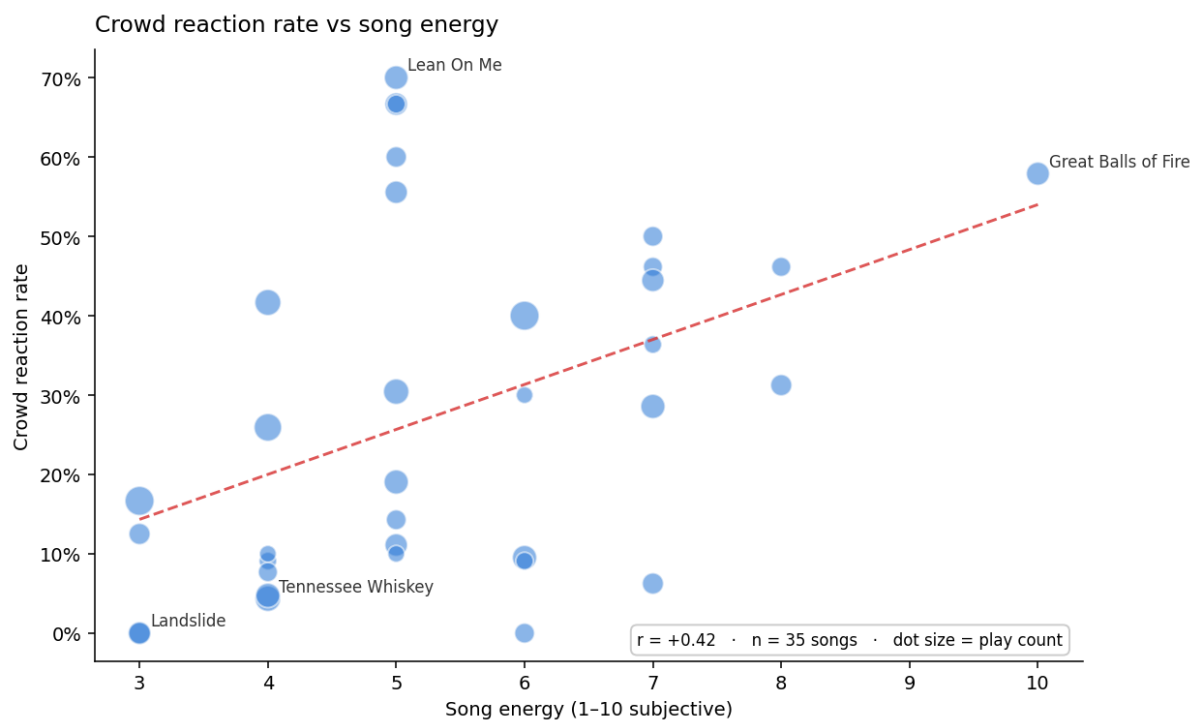


Average tips per play, with and without a crowd reaction

Across all 1,047 plays, ones with a crowd reaction averaged 0.45 tips against 0.33 without ($p=0.009$).

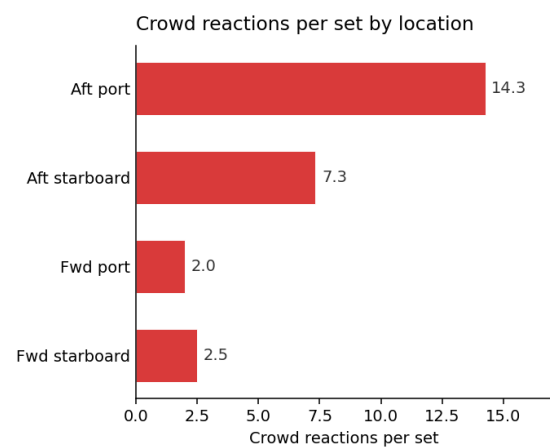
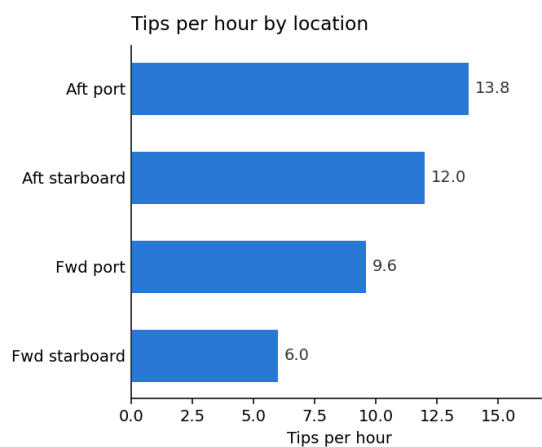
Crowd reaction rate vs song energy

Higher-energy songs draw more crowd reactions. This trend does not carry over to tips — energy predicts singing, not tipping. I scored energy myself, so this is exploratory.



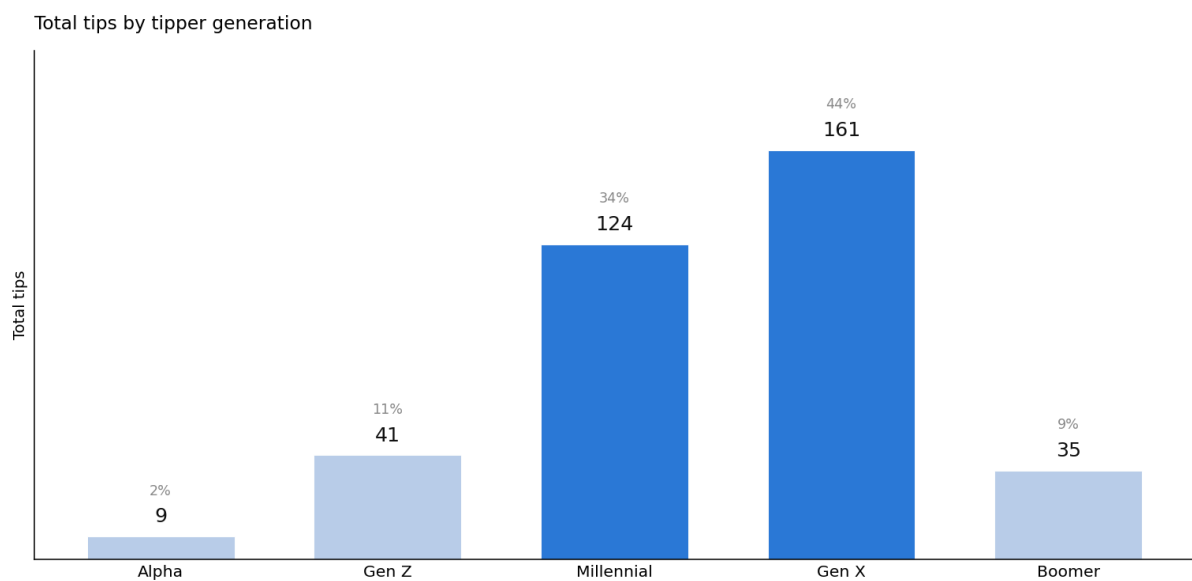
Location

Left: tips per hour by location. Right: crowd reactions per set. Same ranking on both. Aft locations clearly outperform forward ones (12.6 vs 8.4 tips/hour, $p=0.008$). Port trends above starboard on both sides but the sample is too small to be sure. Fwd starboard has just 2 sets — treat as directional. That said, I knew from experience that fwd starboard performed badly which is why it was only used in the experiment a few times.



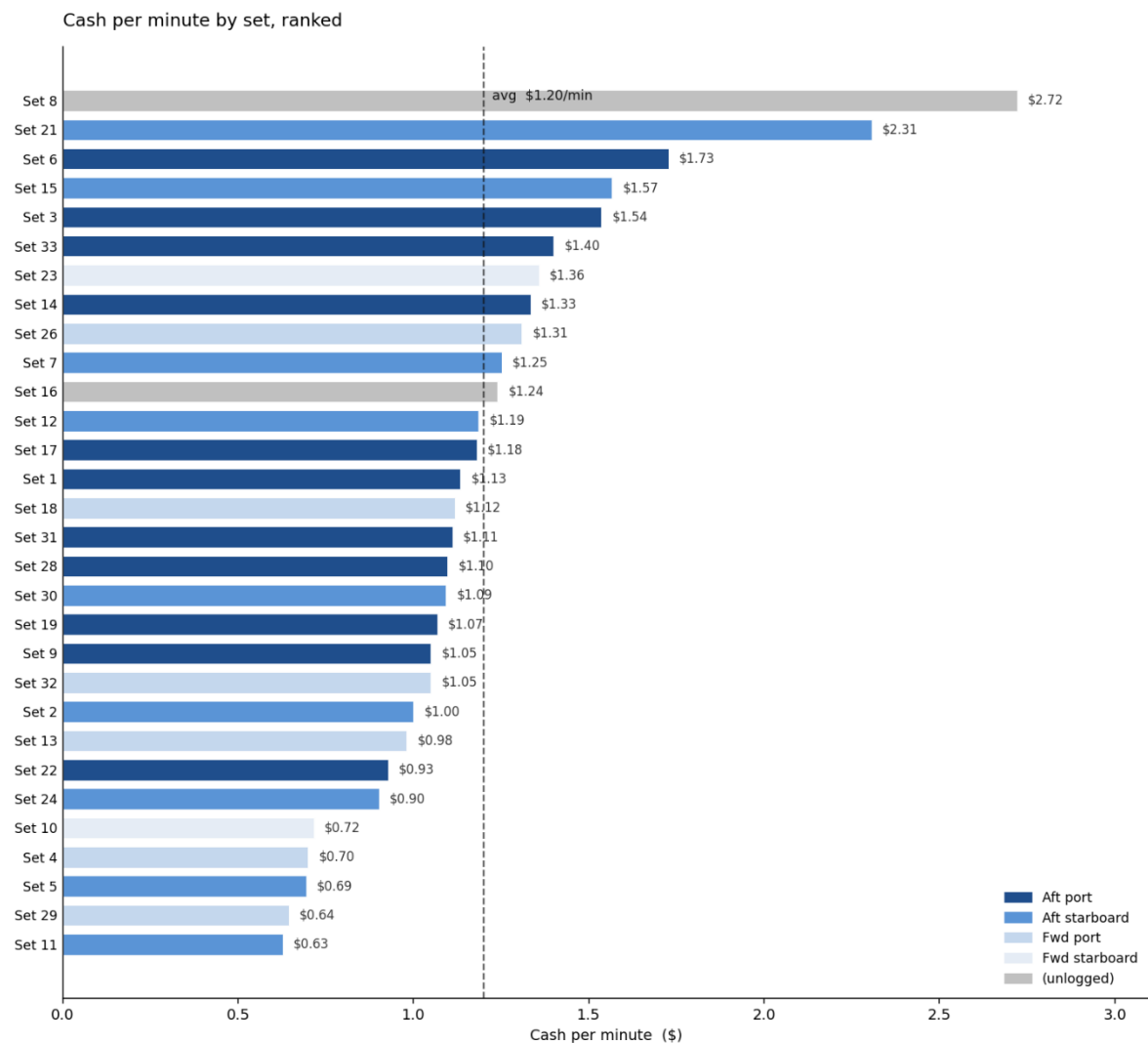
Total tips by tipper generation

Generation of the tipper, my read of the room. Gen X and Millennials together account for 77% of tips. The setlist story downstream is really a "what pays this audience" story — a ship with younger demographics would need the analysis redone.



Cash per minute by set, ranked

All 30 completed sets, sorted by cash rate, coloured by location. Dashed line is the average across sets (\$1.20/min). Note the range — \$0.63 to \$2.72 per minute. This scale of set-to-set variance is why smaller sub-effects (side, hour) are hard to prove on 30 sets: a single set can deviate this much on crowd luck alone.



Strategic recommendations

The clearest thing I've been able to show is that some songs are higher earners than others. I can also show that sets where the crowd sang more were more profitable — though whether the singing drives the tipping, or a good crowd simply does both, the data can't say but my intuition tells me that it is worth every little bit of effort to get people singing in the moment.

It also shows that, in general, songs I know to be crowd pleasers are. I can confidently play the high earners more and drop those that have performed badly. I also checked whether the ship's day port (Nassau, Cococay, Port Canaveral, or a sea day) predicted earnings. It didn't — rates were flat across all four, well within the set-to-set noise.

After years of doing this, nothing is really new to me here. There are a few songs that have performed far better than I expected and some that I can confidently remove from the set. Songs that work well in a piano bar setting don't translate to top-ranked songs in the elevator. Tiny Dancer, for example, has proven to be unsingalongable to the cruise ship masses even though I know it to be a classic that piano bar enthusiasts know and love. It's these discrepancies that make this job challenging. Having such a varied range of demographics makes it very hard to know what to play. As the culture of music grows it is increasingly hard to appeal to the masses with classic songs. Young people don't necessarily know songs new or old; old people definitely know old songs but don't always know massive hits that we assume everyone knows. Knowing this, sticking to the piano bar seems the comfortable option, although playing to a sample more representative of the population has been fun.

I would have liked to continue with this experiment and show the recommendations have worked in my favour. The time I had here was limited, and I think the song-level data shows the limitations of song choice as a predictor of tip earnings. The problem here is that my intuition in choosing a song for a specific group may be the very reason the top songs have performed so well. This is a finding in itself — in many cases I should keep doing what I'm doing. If I wanted to test these songs further I could play them more often, to groups of every sort. This would be counterproductive though; there is no point playing 'Take It Easy' to a group of 13-year-olds. They won't be entertained by it, which is my job, and it won't change the fact that they have no money in their pocket.

The crowd reaction findings are perhaps the most interesting, and more subtle than I expected. A song's tendency to get a reaction tells you nothing about how well it tips — the two rankings barely relate ($r=0.05$). But zoom out from the song and the picture changes: a play that gets a reaction earns more in that moment (0.45 vs 0.33 average tips), and a set with more reactions per minute earns more per minute ($r=0.43$). An engaged room tips more at every level — it's just not something any particular song can reliably produce. Many entertainers who do this job have told me to lean on the crowd-pleasers, and playing songs that get the crowd going is exhausting — sometimes fun and completely natural, other times forced. These findings say I shouldn't worry about that. The engagement matters, but it can't be manufactured from a setlist; it comes from reading the room in front of you. I have always maintained that as entertainers we should keep expectations low and take each set as it comes. There are just too many variables out of our control. We should play whatever we feel at the time — play the moment, not yesterday's or last year's. There are some undoubted hero songs in the piano bar repertoire, but response and tips come as a result of us doing our best at the time, not following a set of data-driven rules. If there are some rules, I'd love to know them and be able to show them, but perhaps on this occasion my instinct is good enough.

Closing thoughts

This project set out to answer two questions: does song choice matter, and does crowd reaction correlate to tips. The answers turned out to be yes, but less than I assumed — and yes, but not where I expected to find it. That's a better outcome than confirming what I already believed. Tracking every tip and reaction mid-performance wasn't easy, even with the app doing the heavy lifting, and I probably won't run the experiment again — not least because I'm moving away from music and into a career in data analytics. As a final project bridging the two, though, I couldn't have asked for a better dataset than a piano, an elevator, and a few thousand strangers.
